

# U.S. Electric Vehicle Market Share Analysis — Findings Report

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## Summary

- **EVs are still a small share of the fleet:** 1.24% of the 287 million registered vehicles in the U.S.
- **They are concentrated in California**, which has 13% of U.S. vehicles but 35% of its EVs. Outside California, the EV share falls to 0.92%.
- **The gap between states is wide.** California's EV share (3.41%) is 27 times North Dakota's (0.13%).
- **Where to invest next:** Texas, Ohio, and Pennsylvania. They combine large fleets with below-average EV share, so they are furthest from the national average in number of vehicles.

## 1. Data Cleaning

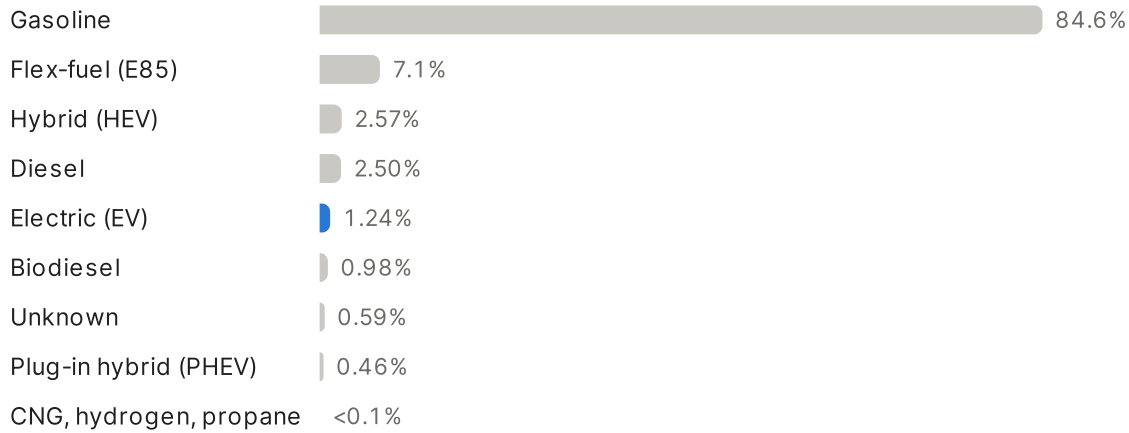
- **Source:** U.S. Department of Energy, Alternative Fuels Data Center, *Vehicle Registration Counts by State*: light-duty vehicles registered in **2023**, based on Experian registration data and rounded to the nearest 100 ([afdc.energy.gov/vehicle-registration](https://afdc.energy.gov/vehicle-registration)).
- **Dataset:** 51 rows (50 states + District of Columbia), 13 columns: state plus 12 fuel types.
- **Fuel types:** Electric (EV), Plug-In Hybrid Electric (PHEV), Hybrid Electric (HEV), Biodiesel, Ethanol/Flex (E85), Compressed Natural Gas (CNG), Propane, Hydrogen, Methanol, Gasoline, Diesel, Unknown Fuel.
- **Data quality issues addressed:** Numeric columns were stored as strings with commas (e.g., "1,256,600"). Removed commas and converted all fuel columns to numbers. No missing values were found. Methanol was zero across all 51 states. State names were already standardized, with no duplicates or inconsistencies.
- **Total registered vehicles in the U.S.:** 287,096,500
- **What "EV share" means here:** battery-electric vehicles as a share of *all registered vehicles* (the cars already on the road), not new-car sales. It changes slowly, so it shows where EV drivers are now rather than where sales are heading.

## 2. Market Share Analysis

### National Fuel Type Breakdown (% of all 287M vehicles)

#### What the U.S. fleet runs on

Share of 287 million registered vehicles by fuel type



| Fuel Type             | % of Total | Vehicle Count |
|-----------------------|------------|---------------|
| Gasoline              | 84.60%     | 242,870,900   |
| Flex-fuel (E85)       | 7.05%      | 20,240,600    |
| Hybrid (HEV)          | 2.57%      | 7,392,300     |
| Diesel                | 2.50%      | 7,184,300     |
| Electric (EV)         | 1.24%      | 3,555,900     |
| Biodiesel             | 0.98%      | 2,803,600     |
| Unknown               | 0.59%      | 1,694,100     |
| Plug-in hybrid (PHEV) | 0.46%      | 1,307,200     |
| CNG                   | 0.009%     | 24,700        |
| Hydrogen              | 0.006%     | 16,900        |
| Propane               | 0.002%     | 6,000         |
| Methanol              | 0.000%     | 0             |

## Top 5 States by EV Share

EVs as a percentage of each state's total registered vehicles.

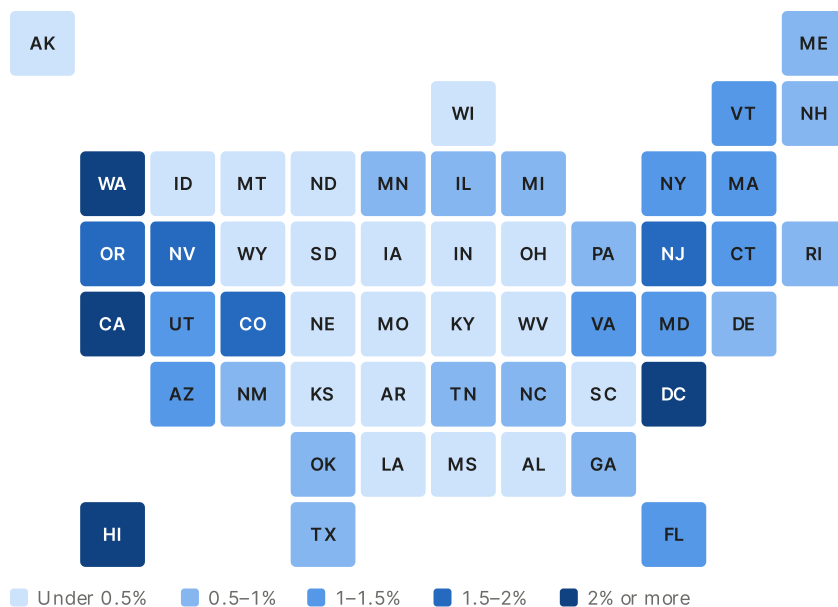
| Rank | State                | EV %  | EV Count  | Total Vehicles |
|------|----------------------|-------|-----------|----------------|
| 1    | California           | 3.41% | 1,256,600 | 36,850,300     |
| 2    | District of Columbia | 2.60% | 8,100     | 312,100        |
| 3    | Hawaii               | 2.37% | 25,600    | 1,080,100      |
| 4    | Washington           | 2.23% | 152,100   | 6,816,100      |
| 5    | Nevada               | 1.85% | 47,400    | 2,556,100      |

## Bottom 5 States by EV Share

| Rank | State         | EV %  | EV Count | Total Vehicles |
|------|---------------|-------|----------|----------------|
| 47   | West Virginia | 0.19% | 2,800    | 1,510,900      |
| 48   | South Dakota  | 0.19% | 1,700    | 918,000        |
| 49   | Wyoming       | 0.17% | 1,100    | 652,900        |
| 50   | Mississippi   | 0.13% | 3,600    | 2,723,300      |
| 51   | North Dakota  | 0.13% | 1,000    | 797,400        |

## EV share of registered vehicles by state

Each square is a state (plus D.C.), placed roughly where it sits on the map



## California vs. Other Large States

| State      | EV %  | PHEV % | HEV % | Gasoline % |
|------------|-------|--------|-------|------------|
| California | 3.41% | 1.11%  | 4.62% | 84.64%     |
| Florida    | 1.37% | 0.31%  | 2.23% | 86.83%     |
| New York   | 1.16% | 0.82%  | 2.77% | 87.52%     |
| Texas      | 0.89% | 0.21%  | 1.73% | 82.93%     |

💡 California leads all large states by a wide margin. Its EV share (3.41%) is 2.5× Florida's, 2.9× New York's, and 3.8× Texas's. California also leads in PHEV and HEV share, so its gasoline share is lower than New York's or Florida's despite having the largest fleet.

## 3. Trend & Insights

### Alternative Fuels: Significant vs. Niche

The three alternative fuels in scope (biodiesel, ethanol, and hydrogen) were evaluated as a percentage of total registered vehicles nationally.

| Fuel               | % of Total | Vehicle Count | Classification                                                                                                                                                  |
|--------------------|------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethanol/Flex (E85) | 7.05%      | 20,240,600    | <b>Largest by count, but overstated.</b> This counts vehicles that <i>can</i> run on E85; most fill up with regular gasoline, so real ethanol use is far lower. |
| Biodiesel          | 0.98%      | 2,803,600     | <b>Modest.</b> Comparable in scale to EVs (1.24%); not negligible in absolute terms.                                                                            |
| Hydrogen           | 0.006%     | 16,900        | <b>Niche.</b> All 16,900 are in California; negligible nationally.                                                                                              |

For context, CNG (0.009%), propane (0.002%), and methanol (0%) are also niche. Together the three fuels in scope are 8.03% of vehicles, almost entirely flex-fuel.

### Key Insights

**How concentrated are EVs?** California has 12.8% of the nation's vehicles but 35.3% of its EVs. Take California out and the EV share of the rest of the country drops from 1.24% to 0.92%.

**Which states are leading?** California, D.C., Hawaii, Washington, and Nevada have the highest EV shares, mostly western and coastal places. The dataset has no charging or policy data, so

explanations like charging access or state incentives are hypotheses to test, not findings.

**Which states lag behind?** North Dakota, Mississippi, Wyoming, South Dakota, and West Virginia are all below 0.2%. They are rural, low-density states where most vehicles run on gasoline or diesel.

**What does this imply for infrastructure planning?** The gap between leading and lagging states is about 27× (California at 3.41% vs. North Dakota at 0.13%). Without targeted investment, lagging states risk falling further behind, creating a two-tier EV landscape.

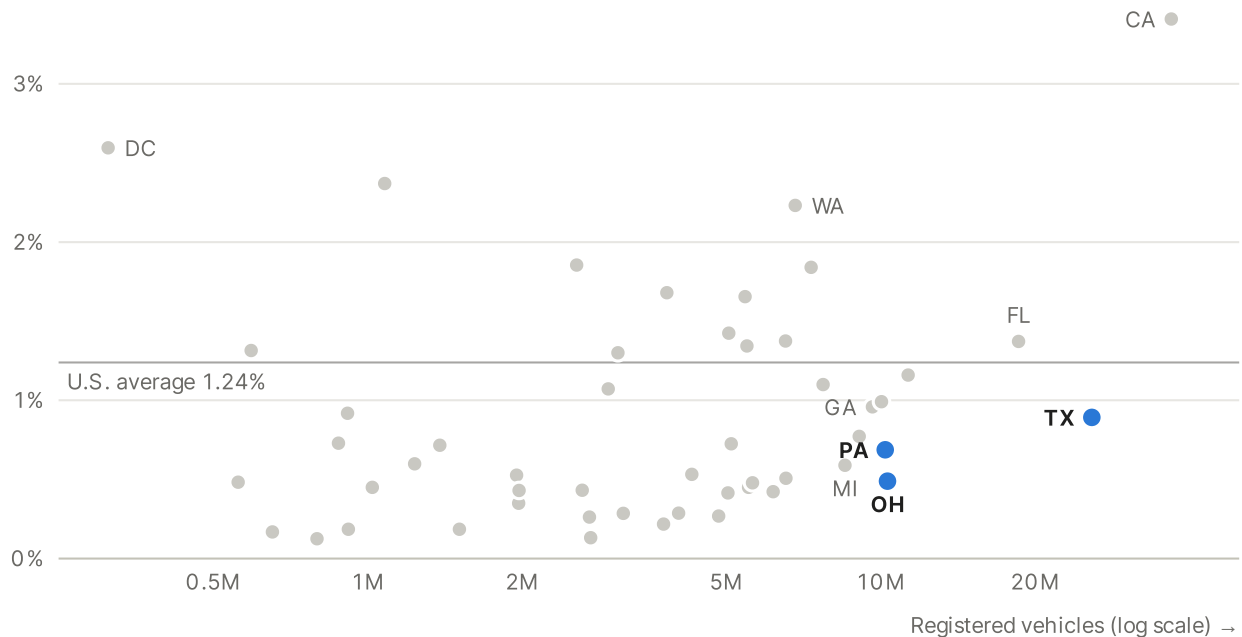
## 4. Recommendations: 3 States to Prioritize for EV Infrastructure Investment

Charging investment reaches the most potential drivers where a state has a lot of vehicles *and* a below-average EV share. To rank states on both at once, I estimated how many more EVs each state would need to reach the national average share:

**EVs short of average** = total vehicles × 1.24% – current EVs

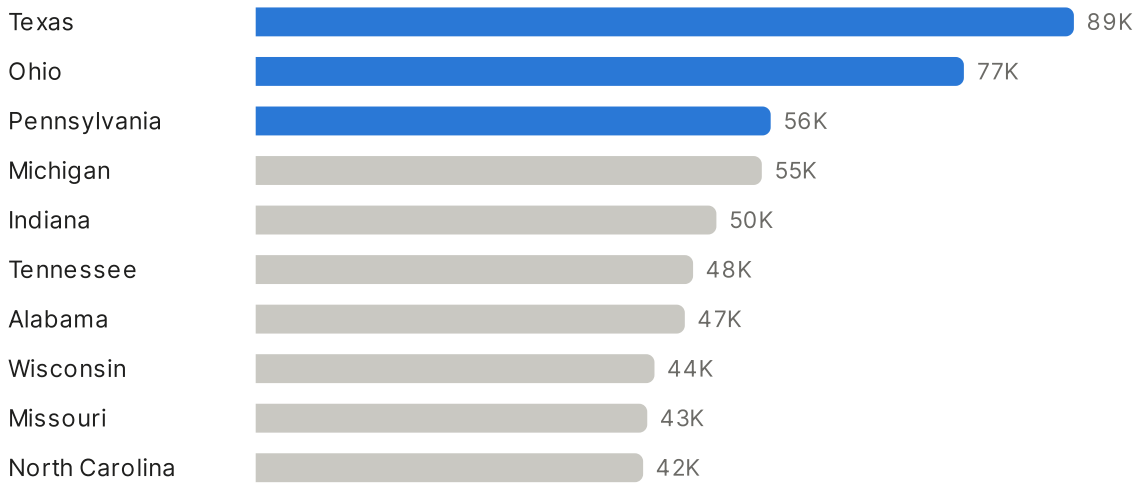
### Big fleets, low EV share

Each dot is a state. Blue dots are the three recommended states.



## EVs short of the national average

How many more EVs each state would need to reach 1.24%. Top 10 states.



### 1. Texas

- **EV share:** 0.89% (below the 1.24% national average)
- **Total fleet:** 25.8M vehicles (#2 in the U.S., 8.99% of all vehicles)
- **EVs short of average:** about 89,000, the largest gap of any state
- **Rationale:** Texas has the second-largest fleet in the country and an EV share well below average. Its gap is the biggest in the country, so infrastructure there would serve the largest base of potential EV drivers; closing even part of it means tens of thousands more EVs on the road.

### 2. Ohio

- **EV share:** 0.49%, less than half the national average
- **Total fleet:** 10.3M vehicles (#5 in the U.S., 3.59% of all vehicles)
- **EVs short of average:** about 77,000
- **Rationale:** Ohio pairs a top-five fleet with the lowest EV share of any of the ten largest fleets, which puts it second only to Texas on this measure.

### 3. Pennsylvania

- **EV share:** 0.69% (below national average)
- **Total fleet:** 10.2M vehicles (#6 in the U.S., 3.56% of all vehicles)
- **EVs short of average:** about 56,000
- **Rationale:** Pennsylvania has almost the same fleet as Ohio and a gasoline share of 87.03%. Michigan (about 55,000 EVs short) is a close fourth and a strong alternative.

## Why not Florida or Georgia?

Both are large states that might seem like obvious picks. **Florida** has the third-largest fleet (18.6M), but its EV share (1.37%) is already above the national average. **Georgia** (0.96%) is a major logistics hub, which could make charging there serve drivers across the Southeast, but it ranks #17 by EVs short of the average.

## 5. Limitations & Next Steps

- This is a 2023 snapshot, so it shows levels, not growth, and EV numbers have grown since. Adding registrations from other years, or new-car sales, would show which states are catching up.
- The gap measure assumes the national average is the right target. Adding public charger counts per state would allow ranking by chargers per EV, which is closer to where investment is actually lacking.

Data, notebooks, and Tableau workbook: [github.com/Mahabala-O/Analyzing-U.S.-Electric-Vehicle-Market-Share](https://github.com/Mahabala-O/Analyzing-U.S.-Electric-Vehicle-Market-Share) · Interactive write-up: [portfolio-mo.vercel.app/projects/us-ev-market-share](https://portfolio-mo.vercel.app/projects/us-ev-market-share)